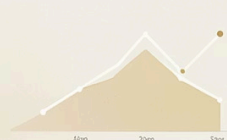


Retail Analytics

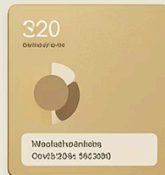


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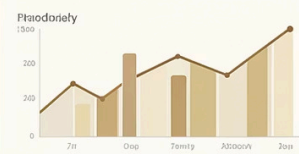
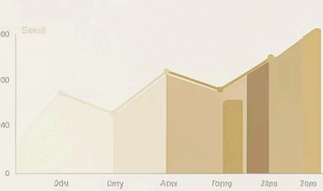
まだ



KKPI Analysis



Gni Petaal Analysis



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

Review ratings imputed

Dataset includes customer demographics, purchase details, and shopping behavior metrics. Key features encompass age, gender, location, subscription status, item purchased, category, purchase amount, season, size, color, discount applied, promo code usage, previous purchases, frequency, review rating, and shipping type.

Data Preparation Process

01

Data Loading

Imported dataset using pandas for analysis

02

Initial Exploration

Checked structure with `df.info()` and summary statistics with `.describe()`

03

Missing Data Handling

Imputed 37 missing review ratings using median by product category

04

Column Standardization

Rewrote columns to snake case for readability

05

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

06

Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame

Revenue by Gender



Key Finding

Male customers generated **\$157,890** in revenue, significantly outpacing female customers at **\$75,191**.

This 2.1x revenue difference suggests gender-based marketing strategies could optimize returns.

Top Products by Rating

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

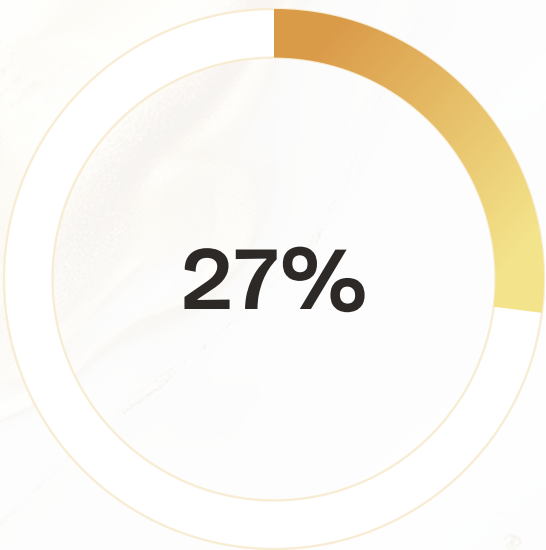
3.78 average rating

These top-rated products represent quality benchmarks and potential campaign highlights.

Shipping & Subscription Analysis

Express Shipping

\$60.48 average purchase

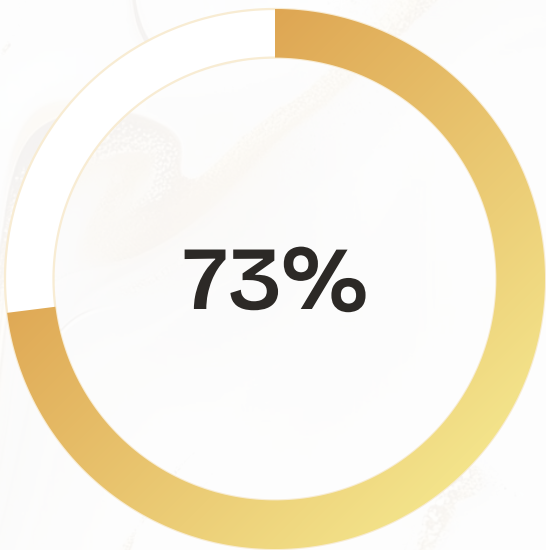


Subscribers

1,053 customers subscribed

Standard Shipping

\$58.46 average purchase



Non-Subscribers

2,847 customers not subscribed

Express shipping correlates with higher spending. Subscribers average \$59.49 per purchase versus \$59.87 for non-subscribers, with total subscriber revenue at \$62,645 versus \$170,436 for non-subscribers.

CUSTOMER LOVILITY TIERS



Customer Segmentation



Loyal

3,116 customers with consistent purchase history



Returning

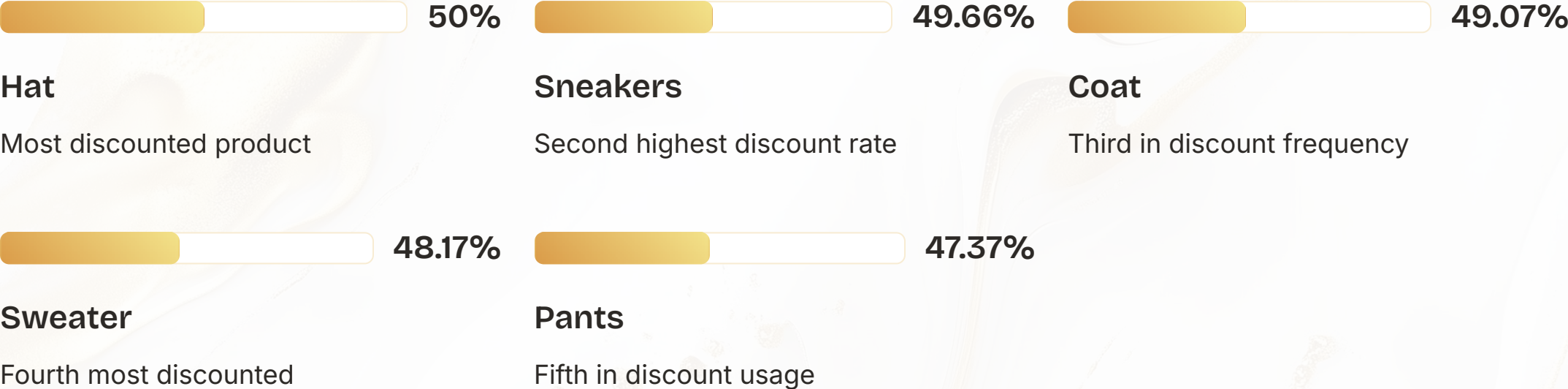
701 customers making repeat purchases



New

83 customers with single transactions

Discount Strategy Insights



These products show highest discount dependency, suggesting careful margin management needed.

Revenue by Age Group

Young Adult

\$62,143 total revenue

Middle-aged

\$59,197 total revenue

Adult

\$55,978 total revenue

Senior

\$55,763 total revenue

Young adults lead in revenue contribution, followed by middle-aged customers. Adult and senior segments show similar performance.

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base



Customer Loyalty Programs

Reward repeat buyers to move them into loyal segment



Review Discount Policy

Balance sales boosts with margin control



Product Positioning

Highlight top-rated and best-selling products in campaigns



Targeted Marketing

Focus on high-revenue age groups and express-shipping users